

On the Semantics of *ye*, *cai* and *dou* in Mandarin Chinese

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Abstract

The Mandarin Chinese particles *ye* and *cai* have multiple functions: distinguished by the prosodic pattern and the optional presence of immediately preceding *lian* and *zhiyou*, *ye* can serve as *also* or a scalar operator, while *cai* has uses as a temporal marker and a scalar operator. However, bar a few exceptions, *ye* and *cai* do not generate free choice inferences in *wh*-item- and *renhe*-NP-constructions nor in disjunctions. I argue that *ye* and *cai* are alternative-sensitive operators and propose a unified semantics for each of them: just as *dou* is a pre-exhaustification exhaustifier operating on sub-alternatives, *ye* and *cai* are pre-exhaustification exhaustifiers operating on orthogonal alternatives, where orthogonal alternatives of a prejacent are propositions that neither entails nor is entailed by the prejacent. I further assert that the semantic compositions regarding *ye* and *cai* are evaluated under three-valued Kleene logic. Both *ye* and *cai* verify the truth of the prejacent and negate the exhaustification of each orthogonal alternative of the prejacent; the difference is that *ye* presupposes the existence of a true orthogonal alternative, whereas *cai* presupposes its non-existence. The semantic weakening operation available to sub-alternatives for *dou* is also available to orthogonal alternatives for *ye* and *cai* to derive their secondary uses as a scalar operator from their primary uses. In addition to alternatives of individuals, disjunctions and scalar items, I enrich the definition to include alternatives of situations and develop an account for a class of idiomatic situational foci in the scope of *dou* and *ye*, using *zuo meng* ‘make dream’ as a prototypical example. I assert that idiomatic situational foci non-literally denote the least or the most likely situation, subject to constraints from their literal core meanings and pragmatics. If the prejacent is interpreted as less likely to happen in the focused situation than any alternative situations, then the use of *dou* is licensed; however, if the prejacent is interpreted as being more likely to happen in the focused situation than in any alternative situations, then a negation of the prejacent is necessary to meet the licensing conditions of *dou*. Consequently, *zuo meng* is licensed in both positive and negative *dou*- and *ye*-constructions in buletic (‘want’) attitudes, licensed only in negative *dou*- and *ye*-constructions in doxastic attitudes and not licensed at all in ‘demand’ attitudes, leading to its part-time behaviour as a negative polarity item.

Contents

1	Introduction	1
2	<i>Yě</i> 也 and <i>cái</i> 才: cousins of <i>dōu</i> 都?	1
2.1	Background on the semantics of <i>dōu</i>	1
2.2	Initial observations on <i>yě</i> and <i>cái</i>	5
2.3	Semantic theory of <i>yě</i>	7
2.4	Semantic theory of <i>cái</i>	12
2.5	Summary	14
3	Semantic similarities among <i>yě</i> 也, <i>cái</i> 才 and <i>dōu</i> 都	15
3.1	Deriving the denotation of <i>yě</i> from that of <i>dōu</i>	15
3.2	Deriving the secondary uses from the primary ones	17
3.3	Summary	19
4	Are <i>yě</i> 也 and <i>cái</i> 才 free choice items?	21
4.1	Observations: (mostly) not FCIs	21
4.2	Semantic theories of <i>yě</i> and <i>cái</i> , in light of FCI data	25
5	<i>Zuò mèng</i> 做梦: a study of <i>dōu</i> 都 over alternatives of situations	28
5.1	Situational foci and <i>dōu</i>	28
5.2	Observations on <i>zuò mèng</i> : a part-time NPI?	32
5.3	Semantic analysis of <i>zuò mèng</i>	34
5.4	Bonus: genericity and <i>dōu</i>	37
5.5	Summary	40
6	Conclusion	40

List of notations

List of glossing abbreviations

3	third person
ATR	attributive
CLF	classifier
EXCLAM	exclamative
IPFV	imperfective
PFV	perfective
PL	plural
RES	resultative
SG	singular

Glossing legend

<i>shìlì</i>	unstressed constituent
<i>shìlì</i>	stressed constituent
<i>shìlì</i> _[+F]	syntactic goal
<i>shìlì</i> _[+F]	syntactic probe

Hyperlink examples

(1)	linguistic example
Xiang 2020	citation

1 Introduction

The Mandarin Chinese particles *dōu*, *yě* and *cái* are known for their variable functions. Over the years, many studies on *dōu* have propelled research in quantification, plural semantics and distributivity, alternative semantics, free choice, exhaustification theory and more. However, despite several decades of research on *dōu*, the proposals remain incredibly diverse, and a consensus has yet to be reached. Alongside is the perplexing connection between *yě* and *dōu*: they are interchangeable in some occasions but not in others, and even in situations in which they are interchangeable in principle, often a slight change in the prosody could cause a deviation in semantic judgement and felicity. Moreover, other grammatical particles in Mandarin Chinese have similar syntactic or prosodic requirements, but with intricate semantics yet to be exposed, with *cái* being one of them. All these topics deserve scrutiny, which I attempt to achieve in this dissertation.

The dissertation is organised as follows. Section 2 introduces the semantics for *yě* and *cái* through their basic uses. Section 3 draws semantic similarities among usages of *yě*, *cái* and *dōu*. Section 4 explores the possibility of *yě* and *cái* as licensors of free choice items (FCIs) and strengthens their semantics with the new observations. Section 5 identifies that focus alternatives can be situations, and investigates the consequences through a case study of an idiomatic situational focus, *zuò mèng*. Section 6 concludes.

2 *Yě* 也 and *cái* 才: cousins of *dōu* 都?

2.1 Background on the semantics of *dōu*

There is a plethora of research on *dōu*: Lin 1998 is one of the pioneers in giving an account of *dōu* as a universal quantifier over a plural NP, selecting the distributive reading of the VP complement rather than the collective reading, thereby known as a generalised distributivity operator over a plurality cover à la Schwarzschild 1996; Portner 2002 is among the first to establish the synonymy between *dōu* and *even*; Hole 2004 recognises a wide range of uses of *dōu*, including a free choice (FC) interpretation of *wh*-items succeeded by *dōu* and a use of

dōu as a universal quantifier over a set of alternatives; Xiang 2008, in an attempt to address the similarity between *dōu* and *even*, states that *dōu* is a maximality operator that extracts the maximal plural individual from a set of plural covers or alternatives; Cheng and Vicente 2013 expands the analysis of *dōu* as an alternative-sensitive operator to include verb focus in verb-doubling sentences.

Based on previous works, Xiang 2020 unifies three uses of *dōu*: as a quantifier-distributor over plural individuals, as a licenser of FC readings for disjunctions, *wh*-items and *rènhé*- ‘any’-phrases, and as an analogue to *even*. She asserts that *dōu* is an alternative-sensitive operator, i.e., its semantics is dependent on the set of alternatives induced by *dōu*.

Free-choice implicature refers to the effect that a disjunction may sometimes entail all of the disjuncts. Taking the following example from Chierchia and Liao 2015:

- (1) a. *We may hire Mary or Sue.*

(Chierchia and Liao 2015: 6c)

- b. *We may hire Mary and we may hire Sue. (although we may not be allowed to hire both)*

(Chierchia and Liao 2015: 6d)

As (1a) entails (1b), (1a) is said to have a free-choice (FC) effect.

A summary of the semantics of alternatives is as outlined below. Rooth 1985, inspired by the question-answer paradigm, develops the notion of focus alternatives to account for the semantics of *only* and *even*: he asserts that each constituent, apart from its primary denotation, has a corresponding set of alternative denotations which undergoes parallel pointwise semantic composition alongside the main semantics. Unless accompanied by a sentential focus (which often coincides with prosodic stress), the alternative denotation is the same as the primary denotation; however, the element in focus has a set of focus (F-)alternatives that encompasses all individuals of the same semantic type. Some lexical items do not involve alternatives in their denotations, but others do, such as *only* and *even*.

Sauerland 2004 and Chierchia 2006 make the notion of alternatives available to scalar items: for lexical items that imply the existence of a scalar degree, they define the set of scalar (σ -)alternatives as the degree itself. Eligible items carry a [+F] or [+ σ] feature which can be syn-

tactically selected by alternative-sensitive particles. Similarly, Sauerland 2004 and Chierchia 2013 extend the concept of alternatives to disjunctions and quantifiers which are postulated to carry a [+D] feature, accordingly known as domain (D-)alternatives. A summary for the definitions of F-, σ - and D-alternatives are reproduced as follows:

- (2) Alternatives ALT, based on Rooth 1985, Sauerland 2004, Chierchia 2006, Chierchia 2013 and Xiang 2020:

For any lexical item α , the set of alternatives for α , $ALT(\alpha)$, satisfies:

$$ALT(\alpha) = \begin{cases} \{[[\alpha]]\} & \text{if } \alpha \text{ does not carry syntactic features of alternatives} \\ D_{\text{type}([[\alpha]])} & \text{if } \alpha \text{ carries a [+F] feature} \\ S & \text{if } [[\alpha]] \in S \text{ where } S \text{ is a scale (a partially ordered set) and } \alpha \text{ carries a } [+ \sigma] \text{ feature} \\ \{\lambda b \lambda a. a \sqcup b, \lambda b \lambda a. a, \lambda b \lambda a. b\} & \text{if } \alpha \text{ is the disjunctive connector } or \text{ and } \alpha \text{ carries a } [+D] \text{ feature} \\ \{[[\alpha]]^{g[D \rightarrow \cdot]} | g(\cdot) \subseteq g(D)\} & \text{if } \alpha \text{ is a quantifier with a syntactic domain variable } D \text{ and } \alpha \text{ carries a } [+D] \text{ feature} \end{cases}$$

where the join operator $\sqcup$ is defined as follows:

- (3) The join operator $\sqcup$, based on Groenendijk and Stokhof 1989, Partee and Rooth 2012 and Xiang 2020:

For any semantic constituents a, b , $a \sqcup b$ ($'a \text{ join } b'$) satisfies:

$$a \sqcup b = \begin{cases} a \vee b & \text{if } \text{type}(a) = \text{type}(b) = t \\ \lambda x_{\tau}. [a(x) \sqcup b(x)] & \text{if } \text{type}(a) = \text{type}(b) = \langle \tau, \sigma \rangle \text{ for some types } \tau, \sigma \\ \text{undefined} & \text{otherwise} \end{cases}$$

Alternatives of higher constituents follow the canonical rules of semantic composition applied pointwise to the members of the daughter alternatives. For ease of writing, for any linguistic expression α , I will use $ALT(\alpha)$ and $ALT([[\alpha]])$ interchangeably.

Based on the definitions of alternatives mentioned above, Xiang 2020 provides a semantics of *dōu* as an alternative-sensitive operator defined as follows:

(4) The denotation of $d\bar{o}u$, based on Xiang 2020: 36, 48:

$$\begin{aligned}
& [[d\bar{o}u]] \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)). O_{\text{PRE}}(p, \text{ALT}(p))(w) = 1 \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)). \\
&\quad p(w) = 1 \wedge \forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [O(q, \text{ALT}(q))(w) = 0] \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)). \\
&\quad p(w) = 1 \wedge \forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [\neg (q(w) = 1 \wedge \forall r_{\langle s,t \rangle} \in \text{EXCL}(q, \text{ALT}(q)) [r(w) = 0])] \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)). \\
&\quad p(w) = 1 \wedge \forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [q(w) = 1 \rightarrow \neg (\forall r_{\langle s,t \rangle} \in \text{EXCL}(q, \text{ALT}(q)) [r(w) = 0])] \\
&= \underbrace{\lambda p_{\langle s,t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p))}_{\text{non-vacuous SUB-alternatives}}. \\
&\quad \underbrace{p(w) = 1}_{\text{true prejacent}} \wedge \underbrace{\forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [q(w) = 1 \rightarrow \neg (\forall r_{\langle s,t \rangle} \in \text{EXCL}(q, \text{ALT}(q)) [r(w) = 0])]}_{\text{anti-exhaustivity: for each true SUB-alternative ... the exhaustification of which is false.}}
\end{aligned}$$

where the notions of an O-exhaustifier O and a pre-exhaustification (PE-)exhaustifier O_{PRE} are defined as follows:

(5) a. The O-exhaustifier O , based on Chierchia 2013 and Xiang 2020:

For any proposition p and its set of alternatives $\text{ALT}(p)$ (likewise henceforth), $O(p, \text{ALT}(p))$ satisfies:

$$O(p, \text{ALT}(p)) = \lambda w_s. p(w) = 1 \wedge \forall r_{\langle s,t \rangle} \in \text{EXCL}(p, \text{ALT}(p)) [r(w) = 0]$$

b. The pre-exhaustification (PE-)exhaustifier O_{PRE} , based on Chierchia 2013 and Xiang 2020:

$$O_{\text{PRE}}(p, \text{ALT}(p)) = \lambda w_s. p(w) = 1 \wedge \forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [O(q, \text{ALT}(q))(w) = 0]$$

The difference in the semantics of the quantifier-distributor use and the *even*-like use of $d\bar{o}u$ is due to a slight difference in the definitions of excludable (excl-)alternatives EXCL and sub-alternatives SUB in these two cases:

- (6) a. Excludable (excl-)alternatives EXCL and sub-alternatives SUB , for the quantifier-distributor use of $d\bar{o}u$:

$$\text{EXCL}(p, \text{ALT}(p)) = \{q_{\langle s, t \rangle} \in \text{ALT}(p) | p \not\subseteq q\}$$

$$\text{SUB}(p, \text{ALT}(p)) = \text{ALT}(p) - \{p\} - \text{EXCL}(p, \text{ALT}(p)) = \{q_{\langle s, t \rangle} \in \text{ALT}(p) | p \subset q\}$$

(Here propositions p, q are viewed as sets of possible worlds in which they are satisfied respectively. $p \subseteq q$ in an intensional semantics is equivalent to $p \vdash q$ (' p entails q ') in an extensional semantics.)

- b. Excludable (excl-)alternatives EXCL_μ and sub-alternatives SUB_μ , for the scalar operator use and the *even*-like use of $d\bar{o}u$:

$$\text{EXCL}_\mu(p, \text{ALT}_\mu(p)) = \{q_{\langle s, t \rangle} \in \text{ALT}(p) | p \not\prec_\mu q\} = \{q_{\langle s, t \rangle} \in \text{ALT}(p) | p <_\mu q\}$$

$$\text{SUB}_\mu(p, \text{ALT}_\mu(p)) = \text{ALT}(p) - \{p\} - \text{EXCL}_\mu(p, \text{ALT}(p)) = \{q_{\langle s, t \rangle} \in \text{ALT}(p) | p >_\mu q\}$$

for some contextually appropriate scale μ provided explicitly or implicitly through the discursive context.

Like $d\bar{o}u$, *yě* has multiple functions in Mandarin Chinese. Compared to papers of $d\bar{o}u$, research of *yě* is relatively impoverished. Both Hole 2004 and Xiang 2008 identify that *yě* bears a resemblance to *also* and has another use exchangeable with $d\bar{o}u$ in its *even*-like use. In the latter use, both studies confirm that *yě* generally has a broader distribution than $d\bar{o}u$. Hole 2004 claims that *yě* is an existential quantifier over domains of alternatives contra $d\bar{o}u$ which is a universal quantifier. As of *cái*, Lai 1999 reveals that it presupposes an event that negates the truth condition of the logical prejacent, with the condition that the negation happens no earlier than the reference time; Hole 2004 identifies *cái* as a negated existential quantifier over the set of alternatives.

2.2 Initial observations on *yě* and *cái*

With the background theory of *yě* and *cái* in mind, let us look at some data. *yě* has a use parallel to that of *also*, while *cái* has a temporal use synonymous to *just*:

- (7) a. Bob had some apple ...

小艾 也 吃-了 苹果.

Xiǎoài yě chī-le píngguǒ.

Alice YE eat-PFV apple

‘Alice also had some apple.’

- b. 小艾 才 吃-了 苹果.

Xiǎoài cái chī-le píngguǒ.

Alice CAI eat-PFV apple

‘Alice just had some apple.’

Yě can be used interchangeably with *dōu* after a *lián* ‘even’-connective; similarly, *cái* can succeed a *zhǐyǒu* ‘only’-connective:

- (8) There were three people in the team: Alice, Bob and the team leader. It was believed that the team leader would be the least likely to be late. However ...

- a. Not only was Bob late ...

(连) 领队 都/也 迟到-了.

(*lián*) *lǐngduì dōu/yě chídào-le.*

LIAN team.leader DOU/YE be.late-PFV

‘Even the team leader was late.’

(based on Xiang 2020: 75)

- b. Neither Alice nor Bob was late ...

(只有) 领队 才 迟到-了.

(*zhǐyǒu*) *lǐngduì cái chídào-le.*

only team.leader CAI be.late-PFV

‘Only the team leader was late.’

(7a), (7b) and (8a), (8b) differ not only in the semantics but also in the prosodic stress: in (7a) and (7b), the stress is situated on *yě* and *cái*, while in (8a) and (8b), the stress is on a constituent selected by *yě/cái*, here manifested as an immediately preceding NP. In usages like (8), *lián* and *zhǐyǒu* are always optional. I will subsequently omit the brackets for them sometimes.

Like *dōu*, both *yě* and *cái* can be accompanied by an explicit scale in a sentence:

- (9) a. 约翰 一-分 钱 都/也 要.
Yuēhàn yī-fēn qián dōu/yě yào.
 John one-CLF.cent money DOU/YE want

‘John wants money, ANY money.’ (lit., ‘John wants to have even one cent.’)
 (based on Xiang 2020: 91a)

- b. 约翰 一-分 钱 都/也 不 要.
Yuēhàn yī-fēn qián dōu/yě bù yào.
 John one-CLF.cent money DOU/YE not want

‘John does not want ANY money.’ (lit., ‘John does not want to have even one cent.’)

- (10) 小王 八-点 才 来.
Xiǎowáng bā-diǎn cái lái.
 Little.Wang eight-CLF.o’clock CAI come

‘Little Wang only arrived at eight o’clock.’
 (Hole 2004, ch. 2: 2)

In (9a) and (9b), both the positive and the negative versions are permissible.

2.3 Semantic theory of *yě*

So far, we have witnessed some uses of *yě* and *cái*: *yě* can be used as a substitution of *also*, or in place of *dōu* for an *even*-effect; *cái* has a *just*- or *only*-like meaning. I would like to develop a unified semantics for each of *yě* and *cái* as an alternative-sensitive particle, inspired by the treatment of *dōu* in Xiang 2020. I will proceed case by case and build the semantics along the way.

The first case concerns the *also*-like use of *yě* as exhibited in (7a). I posit that like *also*, the alternatives triggered by *yě* must not be in an entailment relation with the main clause in either direction. From now on, I will refer to the main clause minus *dōu*, *yě* or *cái* as the prejacent. Examples are as follows:

- (11) a. *小艾 午饭 吃-了 青苹果, 也 吃-了 苹果_[+F].
 **Xiǎoài wǔfàn chī-le qīngpíngguǒ, yě chī-le píngguǒ_[+F].*
 Alice lunch eat-PFV green.apple YE eat-PFV apple

‘For lunch, Alice had some green apple, and also had some apple.’

- b. *小艾 午饭 吃-了 水果, 也 吃-了 苹果_[+F].
 *Xiǎoài wǔfàn chī-le shuǐguǒ, yě chī-le píngguǒ_[+F].
 Alice lunch eat-PFV fruit YE eat-PFV apple

‘For lunch, Alice had some fruit, and also had some apple.’

In *yě*-constructions, the F-alternative of the *yě*-containing clause can be explicitly expressed with an immediately preceding clause, as in (11a) and (11b). Let $p_{g.apple} = [[\text{Alice had some green apple for lunch}]]$, $p_{apple} = [[\text{Alice had some apple for lunch}]]$, and $p_{fruit} = [[\text{Alice had some fruit for lunch}]]$, so $p_{g.apple} \subseteq p_{apple} \subseteq p_{fruit}$. In (11a), the ungrammaticality is due to the only (true) F-alternative $q_1 = p_{g.apple}$ being logically stronger than the prejacent $p = p_{apple}$; while in (11b), the ungrammaticality is due to the only F-alternative $q_2 = p_{fruit}$ being logically weaker than the prejacent $p = p_{apple}$. In short, *yě* presupposes the existence of an F-alternative true in the situation that neither entails the prejacent nor is entailed by the prejacent. This presupposition can be expressed as follows:

- (12) The non-vacuity presupposition, for the *also*-like use of *yě*:

$$\exists q_{\langle s, t \rangle} \in \text{ALT}(p)[q(w) = 1 \wedge p \not\subseteq q \wedge q \not\subseteq p]$$

I refer to the focus alternatives that neither entail the prejacent nor are entailed by the prejacent as **orthogonal** (orth-)alternatives:

- (13) Orthogonal (orth-)alternatives ORTH, for the *also*-like use of *yě*:

$$\text{ORTH}(p, \text{ALT}(p)) = \{q_{\langle s, t \rangle} \in \text{ALT}(p) | p \not\subseteq q \wedge q \not\subseteq p\}$$

Then the presupposition for *yě* could be revised as follows:

- (14) The non-vacuity presupposition, for the *also*-like use of *yě* (revised version):

$$\exists q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]$$

And the denotation of *yě* for its *also*-like use is posited as follows:

(15) The denotation of *yě*, for the *also*-like use:

$$[[yě]] = \underbrace{\lambda p_{\langle s,t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p))}_{\text{non-vacuous true ORTH-alternatives}} [\underbrace{q(w) = 1}_{\text{true prejacent}}] . \underbrace{p(w) = 1}_{\text{true prejacent}}$$

In the first use case of *yě*, I have explored its synonymy with *also*. In the second case discussed below, I aim to address another prominent use of *yě* as a substitute for *dōu* in its scalar operator (and *even*-like) use, such as in (8a). This use of *yě* differs from the *also*-like use of *yě* in that the prosodic stress falls on the focused constituent rather than *yě* (or *dōu*), and that *lián* can optionally appear in the current use but is not allowed in the *also*-like use.

As always, let us start with some linguistic examples. Xiang 2008 spots that *yě* often lacks a maximal implication that *dōu* has. The maximality of *dōu* is illustrated in the following examples:

- (16) a. 孩子-们 去-了 公园.
 háizi-men qù-le gōngyuán.
 child-PL go-PFV park
 ‘The children went to the park.’
 (Xiang 2008: 30)
- b. 孩子-们 都 去-了 公园.
 háizi-men dōu qù-le gōngyuán.
 child-PL DOU go-PFV park
 ‘The children all went to the park.’
 (Xiang 2008: 31)

(16a) is compatible with a situation where a small number of contextually relevant children did not go to the park, whereas (16b) is not. In response to the contrast between *yě* and *dōu*, Hole 2004 states that *yě* is an existential quantifier over the alternatives, while *dōu* is a universal quantifier. In such a theory, the distribution of *yě* should be strictly broader than that of *dōu*; however, as Xiang 2008 rightly points out, there are situations where *dōu* is preferred than *yě*:

- (17) 连 傻瓜 都/#也 知道 太阳 从 东边 升起.
lián shǎguā dōu/#yě zhīdào tàiyáng cóng dōngbiān shēngqǐ.
 LIAN idiot DOU/YE know sun from east rise
 ‘Even an idiot knows that the sun rises in the east.’
 (based on Xiang 2008: 53)

In (17), *shǎguā* ‘idiot’ provides a strong pragmatic bias that an idiot is less likely to know something than an average person; there is a contextually prominent likelihood scale that *dōu* operates on. If *yě* presupposes the mere existence of a scalar alternative or is simply insensitive to the scale, we should expect the use of *yě* to be felicitous here, which is not the case. Therefore, I posit that in (17), *yě* is sensitive to the scale; it necessitates the **lack** of inference that the scalar alternative is either more or less likely than the prejacent. Here, felicity of (17) would imply that there is no expectation one way or the other whether an idiot is more likely to know the issue than an average person, which is not the case; the salient scale made prominent by the emphatic *shǎguā* makes it infelicitous.

This formulation of *yě* explains why many sentences that permit *dōu* are also compatible with *yě*. In these cases, with the proper context, it is plausible that the attitude holder does not have a particular stance on the likelihood of the events salient in context. Here is such an example:

- (18) 我 只 让 他/她 打扫 房间, 但 他/她 连 饭 都/也 烧-好-了.
wǒ zhǐ ràng tā dǎsǎo fángjiān, dàn tā lián fàn dōu/yě shāo-hǎo-le.
 I only ask 3SG clean room but 3SG LIAN meal DOU/YE cook-RES-PFV
 ‘I only asked them to clean the room, but they even/also cooked the meal.’
 (based on Xiang 2008: 54)

In (18), the *yě* version implies the lack of view on whether they were likely to cook a meal, while the *dōu* version embodies a notion of surprise, that they were less likely to cook the meal. Likewise, in (17), the sentence could be made felicitous by removing the strong pragmatic bias, e.g., in a context where the speaker has a very close friend sometimes colloquially referred to as *shǎguā*, where they happen to know the issue in question and there is no expectation whether they are more likely to know the issue than an alternative person or not.

In the same vein, Hole 2004 mentions a use case of *yě* that “expresses resignation, or the fact that the speaker accepts the things the way they are”, as follows:

- (19) a. 年纪也不小了, 小艾还没有工作.
niánjì yě bù xiǎo-le, Xiǎoài hái méi yǒu gōngzuò.
 age YE not small-PFV Alice still not have job

‘It is simply the way that Alice is not very young any more while still not having a job.’

(based on Hole 2004, ch. 2: 70d)

- b. 年纪都不小了, 小艾还没有工作.
niánjì dōu bù xiǎo-le, Xiǎoài hái méi yǒu gōngzuò.
 age DOU not small-PFV Alice still not have job

‘Alice is already not very young any more, but she still does not have a job.’

In (19b), the use of *dōu* would indicate that Alice is less likely (and therefore less expected) to not have a job at this age than at an alternative younger age; in contrast, (19a) lacks such an inference. A speaker may choose to deliberately utter (19a) rather than (19b) to attenuate the moral judgment that could be implied from the sentence or to express acceptance of the current state of affairs; or if a similar first-person sentence is used to talk about oneself, this could be interpreted as resignation.

One thing that (19a) and (19b) have in common, though, is that they implicitly refer to an alternative situation when Alice was younger and things could turn out to be different. This observation confirms my theory that *yě* operates on a scale which, in (19a), orders likelihood of the prejacent in different possible worlds; *yě* specifies the lack of an order relation, or that the order relation is **undefined** between the prejacent and each alternative. I further posit that when *yě* is in a sentence with an explicit scale like in (9a) and (9b), its semantics is the same, except that the lack of an order relation is on the explicit scale rather than the implicit likelihood scale.

Based on the analysis above, I assert the following non-vacuity presupposition and denotation of *yě* in its scalar operator use:

- (20) The non-vacuity presupposition, for the scalar operator use of *yě*:

$$\exists q_{\langle s, t \rangle} \in \text{ALT}_{\mu}(p)[q(w) = 1 \wedge <_{\mu}(p, q) \text{ not defined}].$$

(21) The denotation of *yě*, for the scalar operator use:

$$[[yě]] = \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{ALT}_\mu(p) [q(w) = 1 \wedge <_\mu(p, q) \text{ not defined}]. p(w) = 1$$

2.4 Semantic theory of *cái*

In the initial observations, we have discovered that the two modes of prosodic stress in *yě*-constructions correspond to two readings of *yě*. As *cái* also permits these two modes of prosodic stress, in this section, I seek to construct two readings of *cái* closely mimicking those of *yě*.

The first case stems from Hole 2004's report of the temporal use of *cái* as in (7b). Based on the effect of *cái* in (7b), it may be suspected that *cái* is an aspectual marker. However, *cái* differs from other aspectual markers in Mandarin Chinese: for example, *cái* could be used in the future where *gānggāng* 'just' cannot, and the location of the prosodic stress is also different:

- (22) a. 老王 昨天 刚刚 搬-走。
Lǎowáng zuótiān gānggāng bān-zǒu.
 Old.Wang yesterday just move-RES.away
 'Old Wang has just moved away yesterday.'
- b. 老王 昨天 才 搬-走。
Lǎowáng zuótiān cái bān-zǒu.
 Old.Wang yesterday CAI move-RES.away
 'Old Wang only moved away yesterday.'
- c. *老王 明天 刚刚 要 搬-走。
 **Lǎowáng míngtiān gānggāng yào bān-zǒu.*
 Old.Wang tomorrow just will move-RES.away
 'Old Wang will not have moved away until tomorrow.'
- d. 老王 明天 才 要 搬-走。
Lǎowáng míngtiān cái yào bān-zǒu.
 Old.Wang tomorrow CAI will move-RES.away
 'Old Wang will only move away by tomorrow.'

Given that the examples above indicate that *cái* is not a genuine aspectual particle, I intend to seek another route. (22b) and (22d) possess the presupposition that in none of the days before

yesterday/tomorrow has Old Wang moved away. As moving away on any day neither entails nor is entailed by moving away on any other day, this demonstrates that *cái* necessitates the falsity of all orth-alternatives. This semantic effect can be achieved by just adding a single negation to the non-vacuity presupposition of the *also*-like use of *yě*. Therefore, I posit the denotation of *cái* in its temporal use as follows:

(23) The denotation of *cái*, for the temporal use:

$$[[cái]] = \underbrace{\lambda p_{\langle s,t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\neg \exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]}_{\text{universal false ORTH-alternatives}} \cdot \underbrace{p(w) = 1}_{\text{true prejacent}}$$

Let us now move to the second case of *cái* of interest. *Cái* has an *only*-like use which is akin to the scalar operator use of *yě*, as (8a) and (8b) have demonstrated (repeated below):

(24) There were three people in the team: Alice, Bob and the team leader. It was believed that the team leader would be the least likely to be late. However ...

a. Not only was Bob late ...

(连) 领队 也 迟到-了.

(lián) *lǐngduì* yě chídào-le.

LIAN team.leader YE be.late-PFV

‘Even the team leader was late.’

(based on Xiang 2020: 75, repeated from (8a))

b. Neither Alice nor Bob was late ...

(只有) 领队 才 迟到-了.

(zhǐyǒu) *lǐngduì* cái chídào-le.

only team.leader CAI be.late-PFV

‘Only the team leader was late.’

(repeated from (8b))

Compared to (24a), (24b) has the same prejacent. Still, the presuppositions have opposite truth conditions: in (24a), there exists one true proposition in the eligible alternatives, while in (24b), neither of the propositions in the eligible alternatives are satisfied in the context. Given that the prosodic stress is on the focused constituent rather than on *cái*, I postulate that this use of *cái* is a scalar operator on the alternatives for each of which the order relation between it and

the prejacent is undefined, with the same truth conditions as *yě* apart from an extra negation on the non-vacuity presupposition. This assertion brings the scalar operator *yě-cái* contrast on par with the *also*-temporal *yě-cái* contrast. The argument is further supported by the fact that *cái* can operate on explicit scales, as (10) has demonstrated.

Based on the observations above, a denotation of the scalar marker use of *cái* is posited as follows:

(25) The denotation of *cái*, for the scalar operator use:

$$[[cái]] = \lambda p_{\langle s,t \rangle} \lambda w_s : \neg \exists q_{\langle s,t \rangle} \in \text{ALT}_\mu(p)[q(w) = 1 \wedge <_\mu(p, q) \text{ not defined}]. p(w) = 1$$

2.5 Summary

A summary of the results from this section is displayed in the following matrix:

(26) Summary of denotations of *yě* and *cái*:

$$[[yě/cái]] = \lambda p_{\langle s,t \rangle} \lambda w_s : \text{PRES}[p, w]. p(w) = 1$$

where the non-vacuity presupposition $\text{PRES}[p, w]$ is defined as:

	<i>also</i> -like use/temporal use	scalar operator use
<i>yě</i>	$\exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]$	$\exists q_{\langle s,t \rangle} \in \text{ALT}_\mu(p)[q(w) = 1 \wedge <_\mu(p, q) \text{ not defined}].$
<i>cái</i>	$\neg \exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]$	$\neg \exists q_{\langle s,t \rangle} \in \text{ALT}_\mu(p)[q(w) = 1 \wedge <_\mu(p, q) \text{ not defined}].$

As a side note, due to the subtle differences in the two uses of *yě*, in some occasions, the *also*-like use and the scalar operator use of *yě* can generate very similar sentences, as the following minimal pair demonstrates:

(27) There were three people in the team: Alice, Bob and the team leader. It was believed that the team leader would be the least likely to be late. However ...

a. Alice and Bob were late ...

领队 也 迟到-了.

lǐngduì yě chídào-le.

team.leader YE be.late-PFV

‘The team leader was also late.’

- b. Not only were Alice and Bob late ...

(连) 领队 也 迟到-了.

(lián) *lǐngduì* yě chídào-le.

LIAN team.leader YE be.late-PFV

‘Even the team leader was late.’

Disregarding the optional *lián*, the only phonetic difference between (27a) and (27b) is in the intonation, while the only semantic difference between them is that (27a) presupposes the lack of a judgement regarding whether the team leader is less or more likely to be late than the alternatives, while (27b) presupposes that the team leader is less likely to be late.

A parallel phenomenon exists for *cái*, where the difference in intonation corresponds to the same subtle contrast in semantics:

- (28) a. 小艾 刚刚 才 吃-了 一个 苹果.

Xiǎoài gānggāng cái chī-le yī-gè píngguǒ.

Alice just CAI eat-PFV one-CLF apple

‘Only just now has Alice had an apple.’

- b. 小艾 刚刚 才 吃-了 一个 苹果.

Xiǎoài gānggāng cái chī-le yī-gè píngguǒ.

Alice just CAI eat-PFV one-CLF apple

‘Alice has just had an apple.’

3 Semantic similarities among *yě* 也, *cái* 才 and *dōu* 都

In the previous section, I have studied the *also*-like and the scalar operator uses of *yě*, as well as the temporal and the scalar operator uses of *cái*. In this section, I aim to link all uses of *yě*, *cái* and *dōu* together; at the end of the section, I will visualise the theoretical connections in an arrow diagram.

3.1 Deriving the denotation of *yě* from that of *dōu*

Due to the empirical similarities between *yě* and *dōu*, it helps to compare their denotations. I observe that the denotation of *yě* can be recovered straightforwardly from the denotation of *dōu*

given in Xiang 2020, simply by replacing the sub-alternatives in the denotation of $dōu$ with the orth-alternatives; $yě$ does not negate the exhaustification of the sub-alternatives, but rather the orth-alternatives. To compute the derivation, start by replicating the denotation of $dōu$ in its quantifier-distributor use:

(29) The denotation of $dōu$, for the quantifier-distributor use, based on Xiang 2020 (repeated):

$$\begin{aligned}
& [[dōu]] \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)). O_{\text{PRE}}(p, \text{ALT}(p))(w) = 1 \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)). \\
&\quad p(w) = 1 \wedge \forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [O(q, \text{ALT}(q))(w) = 0] \\
&= \underbrace{\lambda p_{\langle s,t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\exists q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p))}_{\text{non-vacuous SUB-alternatives}}. \\
&\quad \underbrace{p(w) = 1}_{\text{true prejacent}} \wedge \underbrace{\forall q_{\langle s,t \rangle} \in \text{SUB}(p, \text{ALT}(p)) [q(w) = 1]}_{\text{anti-exhaustivity: for each true SUB-alternative ...}} \rightarrow \underbrace{\exists r_{\langle s,t \rangle} \in \text{EXCL}(q, \text{ALT}(q)) [r(w) = 1]}_{\text{the exhaustification of which is false.}}
\end{aligned}$$

Revise the denotation of $dōu$ by supplementing the non-vacuity presupposition with an extra condition that there exists not just a sub-alternative, but a sub-alternative **true** in the situation; this change is semantically trivial without affecting the truth condition of $dōu$, as all sub-alternatives are logically weaker than the prejacent and are therefore automatically true given the stipulation that the prejacent is true. Once that happens, make the substitution $\text{SUB} \rightarrow \text{ORTH}$ to derive the denotation of $yě$ from that of $dōu$:

(30) The denotation of $yě$, for the *also*-like use (derived):

$$\begin{aligned}
& [[yě]] \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]. O'_{\text{PRE}}(p, \text{ALT}(p))(w) = 1 \\
&= \lambda p_{\langle s,t \rangle} \lambda w_s : \exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]. \\
&\quad p(w) = 1 \wedge \forall q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [O(q, \text{ALT}(q))(w) = 0] \\
&= \underbrace{\lambda p_{\langle s,t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]}_{\text{non-vacuous true ORTH-alternatives}}. \\
&\quad \underbrace{p(w) = 1}_{\text{true prejacent}} \wedge \underbrace{\forall q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]}_{\text{anti-exhaustivity: for each true ORTH-alternative ...}} \rightarrow \underbrace{\exists r_{\langle s,t \rangle} \in \text{EXCL}(q, \text{ALT}(q)) [r(w) = 1]}_{\text{the exhaustification of which is false.}}
\end{aligned}$$

where O'_{PRE} is the new pre-exhaustification exhaustifier over orth-alternatives rather than sub-alternatives:

(31) The pre-exhaustification (PE-)exhaustifier over orthogonal (orth-)alternatives O'_{PRE} :

$$O'_{\text{PRE}}(p, \text{ALT}(p)) = \lambda w_s. p(w) = 1 \wedge \forall q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [O(q, \text{ALT}(q))(w) = 0]$$

Since the anti-exhaustivity requirement for orth-alternatives is trivially true (as the prejacent itself is excludable by any orth-alternative of the prejacent, in keeping with the definitions in (6a) and (13)), the denotation of $yě$ in (30) is identical to that derived earlier in (15). Thus, two options of the semantics of $yě$ are open: one as a PE-exhaustifier and one that is not an exhaustifier.

To judge which one is the better approach, I would like to explore the consequences if $yě$ was not a PE-exhaustifier. I follow Chierchia and Liao 2015 and Xiang 2020 in asserting that all lexical items with an active syntactic feature of alternatives (such as [+F] and [+D]) select either an explicit exhaustifier (like a PE-exhaustifier as in (5b) and (31)) or the default covert O-exhaustifier defined in (5a). Thus, if $yě$ was not a PE-exhaustifier, the default covert O-exhaustification has to take place. However, unlike PE-exhaustification that is semantically trivial in the current use, O-exhaustification of *also*-like $yě$ -constructions would lead to unsolicited changes in semantics: namely, O-exhaustification leads to the falsity of all orth-alternatives, which is equivalent to saying that **only** the prejacent is true; this is directly against the *also*-like use of $yě$, which is the existence of a true orth-alternative. Therefore, I maintain that $yě$ is a pre-exhaustification exhaustifier over orth-alternatives as defined in (30).

3.2 Deriving the secondary uses from the primary ones

Now that the link between the quantifier-distributor use of *dōu* and the *also*-like use of $yě$ has been discovered, I attempt to seek further the origin of the scalar operator use of $yě$. I propose that this is a secondary use derived from the primary *also*-like use of $yě$, inspired by Xiang 2020. Xiang 2020 proposes that the *even*-like use of *dōu* is derived from the quantifier-distributor use of *dōu* through a semantic weakening operation: to derive the former from the latter, replace

the measurement for ordering alternatives from logical entailment to likelihood (or another contextually defined scale μ). Under closer inspection, I find that the two uses of *yě* and *cái* are linked by the same pattern: one of them could be derived from the other by the same semantic weakening operation.

In the following, I will demonstrate that the scalar operator use of *yě* can be derived from the *also*-like use; as the denotation of *cái* differs from that of *yě* only by a single negation, the mechanism is precisely the same. Start from repeating the definition of orth-alternatives in the denotation of *yě* in its *also*-like use:

(32) Orthogonal (orth-)alternatives ORTH , for the *also*-like use of *yě* (repeated):

$$\text{ORTH}(p, \text{ALT}(p)) = \{q_{\langle s, t \rangle} \in \text{ALT}(p) \mid p \not\subseteq q \wedge q \not\subseteq p\}$$

Then make the substitution $\subseteq \rightarrow \geq_\mu$:

(33) Orthogonal (orth-)alternatives ORTH_μ , for the scalar operator use of *yě* (derived):

$$\begin{aligned} \text{ORTH}_\mu(p, \text{ALT}_\mu(p)) &= \{q_{\langle s, t \rangle} \in \text{ALT}_\mu(p) \mid p \not\geq_\mu q \wedge q \not\geq_\mu p\} \\ &= \{q_{\langle s, t \rangle} \in \text{ALT}_\mu(p) \mid p \not\leq_\mu q \wedge p \neq_\mu q \wedge p \not\geq_\mu q\} \\ &= \{q_{\langle s, t \rangle} \in \text{ALT}_\mu(p) \mid <_\mu(p, q) \text{ not defined}\} \end{aligned}$$

We obtain the denotation of the scalar operator use of *yě* which is identical to the one produced earlier in (20).

In the current proposal, the only difference between the denotations of *yě* and *dōu* lies in the type of alternatives that the PE-exhaustifier operates on: sub-alternatives for *dōu* and orth-alternatives for *yě*. A mechanism for swapping this parameter is provided by Xiang 2020 when dealing with the FCI-licenser use of *dōu*: she takes advantage of the concept of innocently-excludable (I-excl-)alternatives from Fox 2007:

(34) Innocent excludable (I-excl-)alternatives I-EXCL, based on Fox 2007:

$$\begin{aligned} \text{I-EXCL}(p, \text{ALT}(p)) = & \bigcap \{ S_{\langle st, t \rangle} \subseteq \text{ALT}(p) \mid \underbrace{\{p\} \cap \bigcap \{ \neg q_{\langle s, t \rangle} \mid q \in S \}}_{\text{prejacent is consistent with negation of all alternatives in } S} \neq \emptyset \\ & \wedge \underbrace{\forall S'_{\langle st, t \rangle} \subseteq \text{ALT}(p) \left[\{p\} \cap \bigcap \{ \neg q_{\langle s, t \rangle} \mid q \in S' \} \neq \emptyset \rightarrow S' \not\supseteq S \right]}_{S \text{ is one of the maximal sets in } \text{ALT}(p) \text{ satisfying such condition}} \} \end{aligned}$$

To achieve the desired semantics, Xiang 2020 makes the substitution $\text{EXCL} \rightarrow \text{I-EXCL}$ (and corresponding changes to SUB: $\text{SUB} \rightarrow \text{'I-SUB'}$). Therefore, my assertion of a $\text{SUB} \rightarrow \text{ORTH}$ substitution does not require any more structural complexity in the semantics of *dōu* and *yě* than is already required in Xiang 2020.

Once internal and external theoretical connections for the semantics of *yě* are established, those for *cái* follow naturally, as the denotations differ by only a negation.

3.3 Summary

Synthesising the evidence above, I conclude that both *yě* and *cái* are pre-exhaustification exhaustifiers operating on orthogonal alternatives, which puts them in the same category as *dōu*; they have access to the same weakening operations that produces a scalar operator reading from the primary reading, and all three of them have PE-exhaustifier status which allows them to participate in the semantics of *wh*-item- and *rènhé*-NP-constructions, as I will reveal in the next section. Thus, the denotations of *yě* and *cái* are concluded as follows:

(35) The denotation of *yě*, unified:

$$\begin{aligned} & [[yě]] \\ = & \lambda p_{\langle s, t \rangle} \lambda w_s : \exists q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]. O'_{\text{PRE}}(p, \text{ALT}(p))(w) = 1 \\ = & \underbrace{\lambda p_{\langle s, t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\exists q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]}_{\text{non-vacuous true ORTH-alternatives}}. \\ & \underbrace{p(w) = 1}_{\text{true prejacent}} \wedge \underbrace{\forall q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p)) [q(w) = 1]}_{\text{anti-exhaustivity: for each true ORTH-alternative ...}} \rightarrow \underbrace{\exists r_{\langle s, t \rangle} \in \text{EXCL}(q, \text{ALT}(q)) [r(w) = 1]}_{\text{the exhaustification of which is false.}} \end{aligned}$$

(36) The denotation of *cái*, unified:

$$\begin{aligned}
& [[cái]] \\
& = \lambda p_{\langle s,t \rangle} \lambda w_s : \neg \exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]. \mathcal{O}'_{\text{PRE}}(p, \text{ALT}(p))(w) = 1 \\
& = \underbrace{\lambda p_{\langle s,t \rangle}}_{\text{prejacent}} \underbrace{\lambda w_s}_{\text{world}} : \underbrace{\neg \exists q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]}_{\text{universal false ORTH-alternatives}}. \\
& \quad \underbrace{p(w) = 1}_{\text{true prejacent}} \wedge \underbrace{\forall q_{\langle s,t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]}_{\text{anti-exhaustivity: for each true ORTH-alternative ...}} \rightarrow \underbrace{\exists r_{\langle s,t \rangle} \in \text{EXCL}(q, \text{ALT}(q))[r(w) = 1]}_{\text{the exhaustification of which is false.}}
\end{aligned}$$

where

(37) a. Orthogonal (orth-)alternatives ORTH and excludable (excl-)alternatives EXCL, for the *also*-like use of *yě* and the temporal use of *cái*:

$$\begin{aligned}
\text{ORTH}(p, \text{ALT}(p)) &= \{q_{\langle s,t \rangle} \in \text{ALT}(p) \mid p \not\subseteq q \wedge q \not\subseteq p\} \\
\text{EXCL}(p, \text{ALT}(p)) &= \{q_{\langle s,t \rangle} \in \text{ALT}(p) \mid p \not\subseteq q\}
\end{aligned}$$

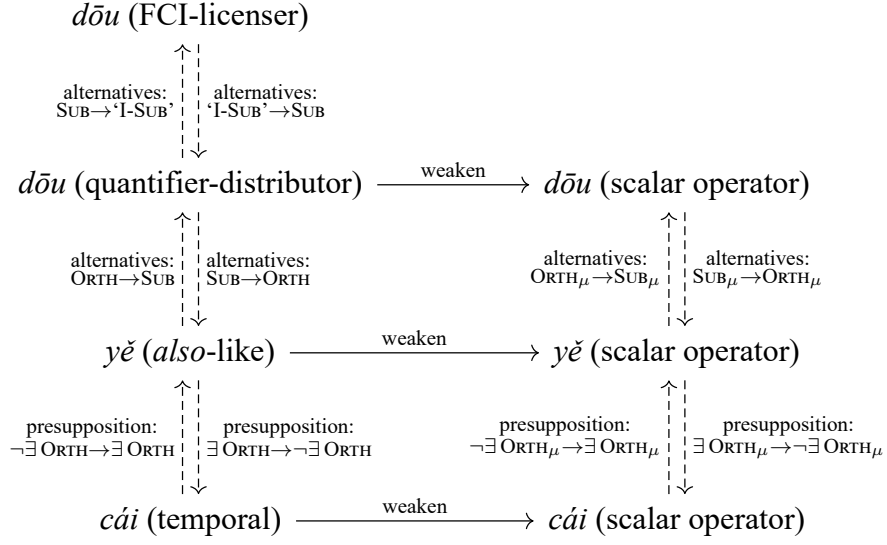
b. Orthogonal (orth-)alternatives ORTH_μ and excludable (excl-)alternatives EXCL_μ , for the scalar operator uses of *yě* and *cái*:

$$\begin{aligned}
\text{ORTH}_\mu(p, \text{ALT}_\mu(p)) &= \{q_{\langle s,t \rangle} \in \text{ALT}(p) \mid <_\mu(p, q) \text{ not defined}\} \\
\text{EXCL}_\mu(p, \text{ALT}_\mu(p)) &= \{q_{\langle s,t \rangle} \in \text{ALT}(p) \mid p <_\mu q\}
\end{aligned}$$

The common core of the semantics of *yě* and *cái* is their presupposition which involves orthogonal alternatives, or alternatives that are **incomparable** to the prejacent. Specifically, the (partial) order relation is undefined between the prejacent and each of the relevant alternatives, with neither larger than the other. Depending on the occasion, this could be $\subseteq$ -incomparability, $\leq_{\text{likely}}$ -incomparability or $\leq_\mu$ -incomparability for some suitably defined scalar $\leq_\mu$ supplied either explicitly or implicitly by the context.

The relationship among uses of *dōu*, *yě* and *cái* are visualised as follows:

(38) Relationship among different uses of *dōu*, *yě* and *cái*:



4 Are *yě* 也 and *cái* 才 free choice items?

According to Xiang 2020, there is a third use of *dōu*, the FCI-licensing use, derived from the primary quantifier-distributor use. In particular, she identifies three types of constructions that could lead to a free choice (FC) inference in Mandarin Chinese: *wh*-items, *rènhé*-NP phrases and disjunctions. She further establishes that the former two have similar distributions and have the same semantics in FC inferences, along the lines of Chierchia and Liao 2015. Given the close links I have just constructed between the semantics of *dōu* and that of *yě* and *cái*, it is profitable to investigate whether *yě* and *cái* have a similar FCI-licensing effect. Hole 2004: 129 recognises that *yě* does not seem to have the same FCI-licensing power as *dōu* does, a phenomenon he cannot explain. Equipped with the results in the previous section, I attempt to inspect the conditions, if any, under which an FC inference is licensed for *yě* and *cái*. The examples will mainly be for *yě*, as *yě* and *cái* do not differ much in this aspect.

4.1 Observations: (mostly) not FCIs

Below are test sentences for FC effects for *wh*-items; *rènhé*-NP phrases conform to the same distribution.

- (39) a. *张三 吃-了 什么.
 **Zhāngsān chī-le shénme.*
 Zhang.San eat-PFV what
 ‘Zhang San has eaten anything.’
 (based on Chierchia and Liao 2015, pp. 1-2)
- b. *张三 可以 买 什么 水果.
 **Zhāngsān kěyǐ mǎi shénme shuǐguǒ.*
 Zhang.San may buy what fruit
 ‘Zhang San may buy any fruit.’
- c. 张三 没 吃 什么.
Zhāngsān méi chī shénme.
 Zhang.San not eat what
 ‘Zhang San has not eaten anything.’
- d. *哪-个 学生 都 进来-了.
 **nǎ-gè xuéshēng dōu jìnlái-le.*
 which-CLF student DOU enter-PFV
 ‘Every student entered.’
 (Xiang 2008: 32)
- e. 哪-个 学生 都 可以 进来.
nǎ-gè xuéshēng dōu kěyǐ jìnlái.
 which-CLF student DOU may enter
 ‘Any student may enter.’
 (Xiang 2008: 33)
- f. 谁 都/*也 知道 这-件 事.
*shuí dōu/*yě zhīdào zhè-jìàn shì.*
 who DOU/YE know this-CLF matter
 ‘Everyone knows about this matter.’
 (Hole 2004, ch. 4: 129)
- g. 谁 都/也 不 会 怪 你.
shuí dōu/yě bù huì guài nǐ.
 who DOU/YE not will blame you
 ‘No one will blame you.’
 (Hole 2004, ch. 3: 78a)

While negation (39c) or the co-presence of *dōu* and a modal operator (39e) license *wh*-items

and *rènhé*-NP phrases and result in an FC effect, *yě/dōu* (39d) or a modal operator (39b) alone does not license *wh*-items. *Yě* licenses a *wh*-item for an FC reading in negative sentences (39g) but not positive sentences (39f).

Note that in (39c), the *wh*-item has to be within the scope of negation to work; if the subject is a *wh*-item instead of the object, the sentence is ungrammatical; however, if a subject *wh*-item is operated by *yě*, an FC reading is available, as the following examples demonstrate:

- (40) a. *谁 没 怪 你.
 **shuí méi guài nǐ.*
 who not blame you
 ‘No one blamed you.’
 b. 谁 也 没 怪 你.
 shuí yě méi guài nǐ.
 who YE not blame you
 ‘No one blamed you.’

As a caveat, although (39a) and (39b) could be rescued by attaching a classifier to the verb (as attested by Chierchia and Liao 2015: 57), I interpret the resulting sentence with an existential reading **without** an FC effect, contrary to what Chierchia and Liao 2015 finds; besides, such a classifier-*wh*-item combination has the option of moving to the top of the sentence under a *yǒu*-‘there be’-construction that *rènhé*-NP phrases does not. These phenomena are demonstrated in the following examples:

- (41) Context: Alice is the admissions officer of a university. After interviewing several students, she decided that some were not up to standards, but the rest were further considered for admission. So among the students that have been interviewed ...
 a. 小艾 可能 会 录取-个 谁.
 Xiǎoài kěnéng huì lùqǔ-gè shuí.
 Alice possibly will admit-CLF who
 ‘Alice could admit someone.’
 b. 有-个 谁 小艾 可能 会 录取.
 yǒu-gè shuí Xiǎoài kěnéng huì lùqǔ.
 exist-CLF who Alice possibly will admit
 ‘There is someone Alice could admit.’

- c. #小艾 可能 会 录取 任何 (一-个) 人.
 #*Xiǎoài kěnéng huì lùqǔ rènhé (yī-gè) rén.*
 Alice possibly will admit any one-CLF person
 ‘Alice could admit anyone.’
- d. *有 任何 (一-个) 人 小艾 可能 会 录取.
 **yǒu rènhé (yī-gè) rén Xiǎoài kěnéng huì lùqǔ.*
 exist any one-CLF person Alice possibly will admit
 ‘There is anyone Alice could admit.’

This use of a *wh*-item following a classifier is only available in a positive sentence but not in a negative one or when interacting with *dōu* or *yě*. That (41c) permits an FC reading that (41a) does not, as well as the discrepancy in the availability of movement to the complement of *yǒu*, imply that this particular instance of *wh*-item deviates from its FC instances and should be treated separately. I posit that CLF is a contracted form of an indefinite determiner (*yī*)-CLF ‘a/some’ which existentially binds the *wh*-item, in turn eliminating the FC reading. I leave the details of this case to further research.

Let us now draw our attention back to the FCI-licensing capabilities of *yě*. From the previous examples, it appears that *yě* does not license *wh*-items as FCIs, except in negations. Further evidence suggests that nor do *yě* support a disjunctive-FCI reading like *dōu* does, as demonstrated in the following minimal examples:

(42) For the next season of the English Premier League ...

- a. 曼城 或者 曼联 可能 会 取胜.
Mànchéng huòzhě Mǎnlián kěnéng huì qǔshèng.
 Man.City or Man.United possibly will win
 ‘Man City or Man United could win.’
- b. #曼城 或者_[+D] 曼联 都_[+D] 可能 会 取胜.
 #*Mànchéng huòzhě_[+D] Mǎnlián dōu_[+D] kěnéng huì qǔshèng.*
 Man.City or Man.United DOU possibly will win
 ‘Both Man City and Man United could win.’

As there could be only one winner of the Premier League, the context only supports an exclusive disjunctive reading but not an FC reading, which leads to the infelicity of (42b).

Unlike *dōu*, *yě* does not appear to interact with the [+D] feature of *huòzhě* ‘or’, as the following examples demonstrate:

(43) Liverpool could win next year’s Premier League ...

- a. [曼城 或者 曼联]_[+F] 也_[+F] 可能 会 取胜.
 [Mànchéng huòzhě Mǎnlián]_[+F] yě_[+F] kěnéng huì qǔshèng.
 Man.City or Man.United YE possibly will win
 ‘Man City or Man United could also win.’
- b. #[曼城 或者_[+D] 曼联]_[+F] 也_[+F] 都_[+D] 可能 会 取胜.
 #[Mànchéng huòzhě_[+D] Mǎnlián]_[+F] yě_[+F] dōu_[+D] kěnéng huì qǔshèng.
 Man.City or Man.United YE DOU possibly will win
 ‘Both Man City and Man United could also win.’

In (43a), the only reading available with *yě* is an *also* reading in the presence of an appropriate orthogonal F-alternative; no infelicitous FC readings are available. Therefore, the felicity of (43a) indicates that *yě* does not create an FC effect on the disjunction.

As of *cái*, it follows the same pattern as *yě* in its failure to offer an FC reading to a disjunction:

(44) Other teams could not win next year’s Premier League ...

- [曼城 或者 曼联]_[+F] 才_[+F] 可能 会 取胜.
 [Mànchéng huòzhě Mǎnlián]_[+F] cái_[+F] kěnéng huì qǔshèng.
 Man.City or Man.United CAI possibly will win
 ‘Only Man City or Man United could win.’

Additionally, *cái* does not license *wh*-items or *rèn hé*-NP phrases as NPIs, with no exceptions that I am aware of.

4.2 Semantic theories of *yě* and *cái*, in light of FCI data

The previous observations find that *yě* and *cái* do not license FC inferences, apart from *yě* in negative *wh*-item sentences. I will address the general trend first, then talk about the exception.

Despite my assertion in the previous section that both *yě* and *cái* are PE-exhaustifiers which, like *dōu*, permits FC readings in principle, the unavailability of FC readings can arise indepen-

dently when the non-vacuity presuppositions of *yě* and *cái* are uniformly unsatisfied in disjunctions and *wh*-item-constructions. The presuppositions are repeated below:

- (45) a. The non-vacuity presupposition for *yě* (repeated):

$$\exists q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]$$

- b. The non-vacuity presupposition for *cái* (repeated):

$$\neg \exists q_{\langle s, t \rangle} \in \text{ORTH}(p, \text{ALT}(p))[q(w) = 1]$$

It is indeed the case for *yě* that the presupposition is rejected in disjunctions and *wh*-item-constructions: a disjunction has no orth-alternatives in its D-alternatives at all, let alone **true** orth-alternatives, as D-alternatives of a disjunction are its disjuncts which always entail the disjunction. However, the presupposition for *cái*, which is a negation to that for *yě*, is vacuously true in these disjunctive environments. Therefore, only one of the presuppositions is false, which fails to explain why *cái* leads to the lack of FC inferences.

However, when disposing of the assumption that classical logic underlies the semantic composition of natural languages, the dissatisfaction of both presuppositions can be achieved: as both presuppositions involve quantification over the set of orth-alternatives which is vacuous in the environments as mentioned earlier, they would both have an **undefined** truth value under a many-valued logic equipped with such a truth value. The simplest among them is the three-valued Kleene logic outlined in Kleene 1938, which I assume here. Under a Kleene-based semantics, an undefined presupposition is unsatisfied and blocks further semantic comprehension, leading to ungrammaticality of *yě* and *cái* in disjunctions and *wh*-item-constructions.

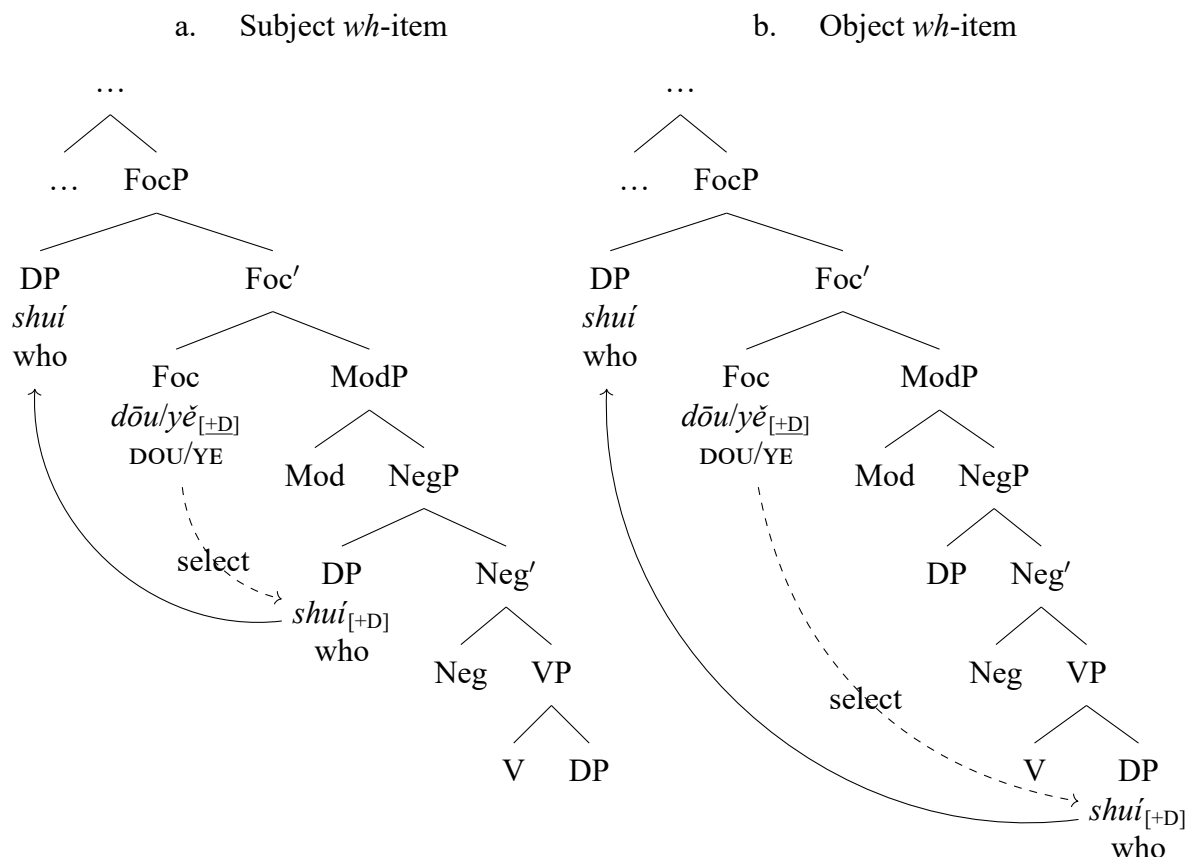
Since we have developed an account for the general FCI-licensing inability of *yě* and *cái*, let us investigate the exceptional case in (39g), where *yě* unexpectedly gives rise to an FC effect in a negative *wh*-item sentence. To interpret this exception in the context, note that (39c), which is the *yě*-less version of (39g), is also unexpectedly grammatical: here, O-exhaustification on *wh*-items should have led to a contradiction. Although PE-exhaustification cannot explain the grammaticality of (39g), O-exhaustification cannot explain the grammaticality of (39c) either. Adding to the fact that such FC effect of *yě* is only available for *wh*-items but not *rènhé*-NP

phrases or disjunctions, I conclude that there is an independent problem with the theory of *wh*-items in Mandarin Chinese in the context of the exhaustification theory, the fixing of which is the prerequisite of solving the exceptional distribution of *yě*.

Although I do not have an alternative theory for *wh*-items in Mandarin Chinese, some light could be shed on the negative *yě* case by noticing the scope restrictions of such FC reading as demonstrated in (39c) and (40a): the *wh*-item has to be within the scope of negation for an FC inference. Therefore, it may help to look for resolutions from a syntactic point of view.

Provided that *yě* and *dōu* tend to occupy the same syntactic positions and are sometimes interchangeable, Shyu 1995, Constant and Gu 2010, etc., have posited that *dōu* and *yě* share a common syntactic framework. Following the syntax of *dōu* and *yě* by Constant and Gu 2010: 18, Xiang 2020: 75 and others, I treat *dōu* and *yě* as heads of a Focus Phrase (FocP); the DP operated by *dōu* undergoes movement to the Spec-FocP position. Their syntactic structures are plotted as follows:

(46) The syntaxes of *dōu* and *yě*, based on Constant and Gu 2010: 18 and Xiang 2020: 75:



Based on the observations above, I assert that *wh*-items in Mandarin Chinese can move in the PF but not in the LF. Therefore, in the LF, a *wh*-item is in the scope of *dōu*, *yě*, negation and modals. As observed previously, the combination of *dōu* and a possibility modal, or negation alone, is sufficient to generate an FC reading, so I suggest that a *wh*-item in the scope of either licensing constituent(s) derives the FC reading. This suggestion explains why combining FCI-licensing constituents with non-FCI-licensing ones does not damage the FC reading and why the FC reading for *wh*-items in negative sentences is only available in the object position but not the subject position. As to why *yě* overrides the scope restrictions of a negation in terms of a FC reading, I posit that *yě* disregards the difference in scope by predicating on the constituents within its scope, including *wh*-items and negation. I leave the exploration of this issue to further research.

5 *Zuò mèng* 做梦: a study of *dōu* 都 over alternatives of situations

5.1 Situational foci and *dōu*

In the previous sections, I have outlined the scalar operator uses of *dōu*, *yě* and *cái*. The scale can be either implicit (usually a likelihood scale) or explicit. Notice that the constituent in focus does not have to be a nominal phrase; rather, the operation could be over embedded clauses and adverbials of time and place. I refer to a focus that denotes a situation, often in the form of an embedded clause or an adverbial, as a situational focus; these cases involve evaluating the prejacent in alternative situations. For example, situational foci triggered by *dōu* are available in both the quantifier-distributor use and the scalar operator (or *even*-like) use. Here are such examples:

- (47) a. 不管 [成 与 不成]_[+F], 你 都_[+F] 要 给 我-个 回话.
bùguǎn [chéng yǔ bù chéng]_[+F], nǐ dōu_[+F] yào gěi wǒ-gè huíhuà.
 no.matter succeed and not succeed you DOU must give I-CLF reply

‘No matter whether you are successful or not, you must give me a reply.’

(Hole 2004, ch. 3: 79)

- b. 老李 连 星期天_[+F] 都_[+F] 工作.
Lǎolǐ lián xīngqītiān_[+F] dōu_[+F] gōngzuò.
 Old.Li LIAN Sunday DOU work
 ‘Old Li works even on Sundays.’
 (Hole 2004, ch. 2: 50c)

Apart from *dōu*, *yě* and *cái* also support situational foci, as the following examples show:

- (48) a. 我 住 在 北京_[+F] 的 时 候, 他/她 也_[+F] 不 跟 我 住 在 一 起.
wǒ zhù zài Běijīng_[+F] de shíhòu, tā yě_[+F] bù gēn wǒ zhù zài yìqǐ.
 I live at Beijing ATR time 3SG YE not with I live at together
 ‘Even when I lived in Beijing, they did not live together with me.’
 (Hole 2004, ch. 2: 61a)
- b. 只有 小王_[+F] 来 我 才_[+F] 去.
zhǐyǒu Xiǎowáng_[+F] lái wǒ cái_[+F] qù.
 only Little.Wang come I CAI go
 ‘Only if Little Wang comes will I go.’
 (Hole 2004, ch. 2: 13)

The situation could involve events ranging from mundane ones (as above) to unlikely but realistic ones:

- (49) 即使 国王 来, 我 都 不 会 去.
jíshǐ guówáng lái, wǒ dōu bù huì qù.
 even.if king come I DOU not will go
 ‘Even if the king comes, I will not go.’
 (Hole 2004, ch. 2: 49a)

to surreal, idiomatic ones:

- (50) a. 连 傻瓜 都 知道 太阳 从 东边 升起.
lián shǎguā dōu zhīdào tàiyáng cóng dōngbiān shēngqǐ.
 LIAN idiot DOU know sun from east rise
 ‘Even an idiot knows that the sun rises in the east.’
 (based on Xiang 2008: 53, repeated from (17))

- b. 你 没有 上 船, 连 船 的 影子 都 没有 见-到.
nǐ méiyǒu shàng chuán, lián chuán de yǐngzi dōu méiyǒu jiàn-dào.
 you not board boat LIAN boat ATR shadow DOU not see-RES
 ‘You have never been aboard, you have not even seen the shadow of the boat yet.’
 (Hole 2004, ch. 2: 52c)

Here are more examples where idiomatic situational foci are used:

- (51) a. 小艾 连 世界末日 都 要 泡 一杯 茶.
Xiǎoài lián shìjiè mòrì dōu yào pào yī-bēi chá.
 Alice LIAN doomsday DOU must make one-CLF tea
 ‘Alice HAS to make a cup of tea every day.’ (lit., ‘Alice has to make a cup of tea even on doomsday.’)
- b. 小艾 连 做 鬼 都 不 会 饶-过 小白.
Xiǎoài lián zuò guǐ dōu bù huì ráo-guò Xiǎobái.
 Alice LIAN be ghost DOU not would forgive-RES Bob
 ‘Alice would NEVER forgive Bob.’ (lit., ‘Alice would not forgive Bob even if she was a ghost.’)

The truth conditions of (50a), (50b), (51a) and (51b) do not involve the literal evaluation of the likelihood that an idiot knows that the sun rises in the east, the likelihood that somebody sees the shadow of the boat, the likelihood that Alice makes a cup of tea on doomsday, or the likelihood that Alice would forgive Bob when she became a ghost; the focused NPs simply imply the least likely event in a non-literal fashion, concurring with the semantics of *dōu* in presupposing that the prejacent is the least likely in the current situation out of all the situational alternatives. The truth conditions of the sentences are equivalent to when the idiomatic foci are replaced with *rènhé*-NP phrases or *wh*-items, as discussed in the previous section.

Notably, many of these idiomatic situational foci restrict the type of events the prejacent can contain. For example, *yǐngzi* ‘shadow’ only works when the prejacent is about seeing or experiencing something; it would be odd to use it in other circumstances:

- (52) #小艾 不 想念 小白, 连 小白 的 影子 都 不 想念.
#Xiǎoài bù xiǎngniàn Xiǎobái, lián Xiǎobái de yǐngzi dōu bù xiǎngniàn.
 Alice not miss Bob, LIAN Bob ATR shadow DOU not miss
 ‘Alice NEVER misses Bob.’ (lit., ‘Alice does not even miss Bob’s shadow.’)

As the shadow is a visual phenomenon, it is unsurprising that it is primarily associated with visual events, but not necessarily events involving thinking about shadows. Along the same lines, it may be expected that other idiomatic foci also have a licensing condition that is generalised from, yet related to, its literal core meaning.

It is noteworthy that *shǎguā* in (50a) is a positive polarity item (PPI) and *yǐngzi* in (50a) is a negative polarity item (NPI). Meanwhile, *shìjiè mò rì* in (51a) and *zuò guǐ* in (51b) can appear in either positive or negative sentences. The distributions of these situational foci prove to be solely determined by the core meanings of the focused expressions and the pragmatics: a *shǎguā* ‘idiot’, by its lexical meaning, is less likely to know of an issue than an average person, so the prejacent which pertains to the knowledge of an idiot has to be less likely than the alternatives; this means only positive sentences license the use of *dōu*. Similarly, for *yǐngzi* ‘shadow’, it is more likely to see a shadow of an object before seeing any other parts of the object (if ever), so the (unnegated) prejacent always contains a proposition more likely than the alternatives; to license *dōu*, the likelihood order has to be inverted, which explains why *dōu* is only grammatical under negation.

As of *zuò guǐ* ‘be ghost’ and *shìjiè mò rì* ‘doomsday’, they license positive *dōu* when the prejacent is less likely to happen in the focused situation than the alternative situations, while they license negative *dōu* when the prejacent is more likely to happen in the focused situation than the alternative ones. For instance, here are the counterparts of (51a) and (51b) opposite in their order relations:

- (53) a. 小艾 连 世界末日 都 不 愿 回 老家。
Xiǎoài lián shìjiè mò rì dōu bù yuàn huí lǎojiā.
 Alice LIAN doomsday DOU not wish return hometown

‘Alice would NEVER want to return to her hometown.’ (lit., ‘Alice would not return to her hometown even on doomsday.’)

- b. 小艾 连 做 鬼 都 要 赌博。
Xiǎoài lián zuò guǐ dōu yào dǔbó.
 Alice LIAN be ghost DOU must gamble

‘Alice HAS to gamble.’ (lit., ‘Alice has to gamble even if she is a ghost.’)

5.2 Observations on *zuò mèng*: a part-time NPI?

Based on the framework above, let us look at another idiomatic situational focus that is available in *dōu*- and *yě*-constructions: *zuò mèng* ‘make dream’. Pearson and Sode 2019 establishes the non-literal use of *in my wildest dreams* as an NPI, as follows:

- (54) a. *Not in my wildest dreams did I think I would meet Uli.*

(Pearson and Sode 2019: 1)

- b. *#In my wildest dreams I thought I would meet Uli.*

(Pearson and Sode 2019: 3c)

In my unpublished work (Ding 2021), I observe a similar NPI effect for *zuò mèng* in Mandarin Chinese:

- (55) a. 我连做梦都没想到/过 我会见到小艾。
wǒ lián zuò mèng dōu méi xiǎng-dào/guò wǒ huì jiàn-dào Xiǎoài.
 I LIAN make dream DOU not think-RES I would meet-RES Alice

‘I NEVER thought I would meet Alice.’ (lit., Even when I make dreams, I did not think I would meet Alice.)

(based on Ding 2021: 4a)

- b. #我连做梦都想到-了 我会见到小艾。
#wǒ lián zuò mèng dōu xiǎng-dào-le wǒ huì jiàn-dào Xiǎoài.
 I LIAN make dream DOU think-RES-PFV I would meet-RES Alice

‘I have ALWAYS thought I would meet Alice.’ (lit., Even when I make dreams, I thought I would meet Alice.)

It is reported that *in my wildest dreams* is not licensed by attitudes other than a doxastic one:

- (56) *#Not in my wildest dreams did I wish/demand that I would meet Uli.*

(Pearson and Sode 2019: 2a)

However, for *zuò mèng*, there are other propositional attitudes compatible with the non-literal meaning of *zuò mèng*, as manifested below. Note the different meaning of *xiǎng* that departs from the doxastic ‘envisage’ meaning:

- (57) 我 连 做 梦 都 想/希望 见-到 小艾.
wǒ lián zuò mèng dōu xiǎng/xīwàng jiàn-dào Xiǎoài.
 I LIAN make dream DOU want meet-RES Alice

‘I would ALWAYS want to meet Alice.’ (lit., Even when I make dreams, I would want to meet Alice.)

(based on Ding 2021: 2b)

In (57), *xiǎng* and *xīwàng* are buletic (‘want’) attitude verbs. Thus, (57) illustrates that *zuò mèng* licenses buletic attitudes, unlike its English counterpart.

Interestingly, unlike a doxastic attitude, *zuò mèng* is not an NPI when licensed by a buletic attitude:

- (58) 我 连 做 梦 都 不 想/希望 见-到 小艾.
wǒ lián zuò mèng dōu bù xiǎng/xīwàng jiàn-dào Xiǎoài.
 I LIAN make dream DOU not want meet-RES Alice

‘I would NEVER want to meet Alice.’ (lit., Even when I make dreams, I would not want to meet Alice.)

However, *zuò mèng* does not license *yāoqiú* ‘demand’ for either positive or negative cases:

- (59) a. #我 连 做 梦 都 要求 见-到 小艾.
 #*wǒ lián zuò mèng dōu yāoqiú jiàn-dào Xiǎoài.*
 I LIAN make dream DOU demand meet-RES Alice
- b. #我 连 做 梦 都 不/没有 要求 见-到 小艾.
 #*wǒ lián zuò mèng dōu bù/méiyǒu yāoqiú jiàn-dào Xiǎoài.*
 I LIAN make dream DOU not demand meet-RES Alice

‘I would ALWAYS demand to meet Alice.’ (lit., Even when I make dreams, I would demand to meet Alice.)

‘I have/would NEVER demand(ed) to meet Alice.’ (lit., Even when I make dreams, I have/would not demand(ed) to meet Alice.)

In other words, it appears that *zuò mèng* is a part-time NPI: it is an NPI in doxastic attitudes, fully licensed in buletic attitudes and not licensed at all in ‘demand’ attitudes. This puzzling behaviour is worthy of investigation, which I attempt to address in the following subsection.

As a side note, there are two choices of negative particles in Mandarin Chinese, *méi* and *bù*, which have subtle differences in semantics between them. In many cases, only one could be

used in any given sentence, but there are cases where either one could be used. In these cases, *méi* sometimes suggests that the prejacents happened in the past, although it is not always the case. Here is a minimal pair that manifests the difference between them:

- (60) a. 我连做梦都没想到/过我会见-到小艾。
 wǒ lián **zuò mèng** dōu méi xiǎng-dào/guò wǒ huì jiàn-dào Xiǎoài.
 I LIAN make dream DOU MEI.not think-RES I would meet-RES Alice
 ‘I NEVER thought I would meet Alice.’ (lit., Even when I make dreams, I did not think I would meet Alice.)
 (based on Ding 2021: 4a, repeated from (55a))
- b. 我连做梦都想-不到我会见-到小艾。
 wǒ lián **zuò mèng** dōu xiǎng-bù-dào wǒ huì jiàn-dào Xiǎoài.
 I LIAN make dream DOU think-BU.not-RES I would meet-RES Alice
 ‘I would NEVER think I would meet Alice.’ (lit., Even when I make dreams, I would not think I would meet Alice.)
 (based on Ding 2021: 2e)
- c. 我连做梦都不会想到我会见-到小艾。
 wǒ lián **zuò mèng** dōu bù huì xiǎng-dào wǒ huì jiàn-dào Xiǎoài.
 I LIAN make dream DOU BU.not will think-RES I would meet-RES Alice
 ‘I would NEVER think I would meet Alice.’ (lit., Even when I make dreams, I will not think I would meet Alice.)

Compared to (60a), (60b) and (60c) do not imply that the contemplation happened in the past; instead, they take on a subjunctive flavour. It is unclear how the choice of *bù* and *méi* affects the semantics of these sentences, but it seems that the impact is confined to aspect and mood and is not central to the current issue, so I will leave it open for further discussion.

5.3 Semantic analysis of *zuò mèng*

Let us see whether the observations regarding *zuò mèng* follow the framework founded earlier in the section. Both *mèng* and *dream* are incredibly varied in their meaning. In one class of use, *mèng* and *dream* both imply a sense of improbability of a situation or an event, as demonstrated in the following idiomatic uses:

- (61) a. 白日-梦
báirì-mèng
 day-dream
 ‘daydream’
- b. 黄梁-一-梦
huángliáng-yī-mèng
 millet-one-dream
 ‘a pipe dream’ (lit., ‘a millet dream’)
- c. 醉-生-梦-死
zuì-shēng-mèng-sǐ
 intoxicated-live-dream-die
 ‘living in a dream’ (lit., ‘living intoxicated and dying dreaming’)
- d. 你 做 梦 吧!
nǐ zuò mèng ba!
 you make dream EXCLAM
 ‘In your dreams!’

In another class of use, *mèng* and *dream* pertain to the strong desire of an attitude holder:

- (62) a. 一个 梦-寐-以-求 的 工作
yī-gè mèng-mèi-yǐ-qiú de gōngzuò
 one-CLF dream-sleep-therein-seek ATR job
 ‘a dream job’ (lit. ‘a job that is sought in dreams and at sleep’)
- b. 美-梦-成-真
měi-mèng-chéng-zhēn
 nice-dream-become-true
 ‘a dream come true’

Based on the observations above, it can be inferred that *zuò mèng* may select a doxastic or a buletic attitude in its use as an idiomatic situational focus with *dōu*. In its doxastic use, the lexical meaning of *zuò mèng* dictates that the thoughts and beliefs held in a dream are less realistic than those held awake, so it is more likely for an unrealistic belief to be held in a dream than in alternative states; that the (unnegated) preadjacent is more likely to happen in the focused situation than in the alternatives coincides with the circumstance with *yǐngzi* in (50b),

which means that *zuò mèng* can only occur in negative sentences. However, *zuò mèng* in bulietic attitudes is not biased towards positive or negative uses, as hope does not need to coincide with likelihood. As *zuò mèng* has no lexical connotations of demand, it cannot be used as a situational focus of *dōu* in this way. All of the results perfectly predicts the data on the non-literal uses of *zuò mèng*.

To further support my judgements, I have conducted searches on the online BCC Chinese Corpus (Xun et al. 2016), one for the interaction between *zuò mèng* and *dōu*, and the other for the interaction between *zuò mèng* and *yě*. The search for *dōu* yields the following results:

(63) Prevalence of *zuò mèng dōu*-constructions in BCC Chinese Corpus (Xun et al. 2016):

Example No.	Example: 做梦都...	Transcription: <i>zuò mèng dōu</i> ...	Glossing: make dream DOU...	No. of entries	Author's judgement
(55a)	没想-到	<i>méi xiǎng-dào</i>	not envisage-RES	88	✓
(55a)	没想-过	<i>méi xiǎng-guò</i>	not envisage-RES	13	✓
(60b)	想-不-到	<i>xiǎng-bù-dào</i>	envisage-not-RES	99	✓
(60c)	不会想-到	<i>bù huì xiǎng-dào</i>	not will envisage-RES	18	✓
(55b)	想-到	<i>xiǎng-dào</i>	envisage-RES	0	#
(55b)	想-过	<i>xiǎng-guò</i>	envisage-RES	4 [†]	#
(57)	想	<i>xiǎng</i>	want	540	✓
(57)	希望	<i>xīwàng</i>	hope	18	✓
(58)	不想	<i>bù xiǎng</i>	not want	0	✓
(58)	不希望	<i>bù xīwàng</i>	not hope	1	✓
(59a)	要求	<i>yāoqiú</i>	demand	0	#
(59a)	不要求	<i>bù yāoqiú</i>	not demand	0	#
(59b)	没有要求	<i>méiyǒu yāoqiú</i>	not demand	0	#

†: including literal, non-idiomatic uses of *zuò mèng*

As can be observed, theoretically predicted felicity generally correlates to a large number of entries in the corpus, whereas infelicitous cases return few or no entries in the corpus after correcting for literal uses of *zuò mèng*.

As discussed in the previous sections, the licensing condition for *yě* is similar to that of *dōu*. The search for *yě* yielding the following results:

(64) Prevalence of *zuò mèng yě*-constructions in BCC Chinese Corpus (Xun et al. 2016):

Example: 做梦也...	Transcription: <i>zuò mèng yě</i> ...	Glossing: make dream YE...	No. of entries	Author's judgement
没想到	<i>méi xiǎng-dào</i>	not envisage-RES	569	✓
想过	<i>méi xiǎng-guò</i>	not envisage-RES	39	✓
想不到	<i>xiǎng-bù-dào</i>	envisage-not-RES	630	✓
不会想到	<i>bù huì xiǎng-dào</i>	not will envisage-RES	95	✓
想到	<i>xiǎng-dào</i>	envisage-RES	5 [†]	#
想过	<i>xiǎng-guò</i>	envisage-RES	0	#
想	<i>xiǎng</i>	want	709	✓
希望	<i>xīwàng</i>	hope	0	✓
不想	<i>bù xiǎng</i>	not want	4	✓
不希望	<i>bù xīwàng</i>	not hope	0	✓
要求	<i>yāoqiú</i>	demand	0	#
不要求	<i>bù yāoqiú</i>	not demand	0	#
没有要求	<i>méiyǒu yāoqiú</i>	not demand	0	#

†: including literal, non-idiomatic uses of *zuò mèng*

The same trend emerges: plenty of entries exist for felicitous cases, but few for infelicitous ones.

5.4 Bonus: genericity and *dōu*

So far, a theory for situational alternatives has helped explain the distribution of *zuò mèng* in propositional attitudes involving *dōu*. Meanwhile, there are other intriguing occasions in which *dōu* operates on situations. A class of examples is generic sentences, also known as habitual sentences:

- (65) a. 熊 吃 蜂蜜.
xióng chī fēngmì.
 bear eat honey
 ‘Bears eat honey.’

- b. 熊 都 吃 蜂蜜.

xióng dōu chī fēngmì.

bear DOU eat honey

‘All bears eat honey./Bears eat honey all the time.’

- (66) a. 我 乘 公交车 上学.

wǒ chéng gōngjiāochē shàngxué.

I take bus go.to.school

‘I take the bus to school.’

- b. 我 都 乘 公交车 上学.

wǒ dōu chéng gōngjiāochē shàngxué.

I DOU take bus go.to.school

‘I always take the bus to school.’

In each pair, the sentence with *dōu* imposes stricter maximality than the counterpart without *dōu*. A similar effect has been captured in Xiang 2008 for non-generic sentences:

- (67) a. 孩子-们 去-了 公园.

háizi-men qù-le gōngyuán.

child-PL go-PFV park

‘The children went to the park.’

(Xiang 2008: 30, repeated from (16a))

- b. 孩子-们 都 去-了 公园.

háizi-men dōu qù-le gōngyuán.

child-PL DOU go-PFV park

‘The children all went to the park.’

(Xiang 2008: 31, repeated from (16b))

However, the strict maximality effect in (65b) and (66b) can be negated by adding *yìbān* or *tōngcháng* ‘usually’ to the sentences, resulting in sentences that are similar in meaning to (65a) and (66a):

- (68) 熊 一般/通常 都 吃 蜂蜜.

xióng yìbān/tōngcháng dōu chī fēngmì.

bear usually DOU eat honey

‘Bears usually eat honey.’

- (69) 我 一般/通常 都 乘 公交车 上学.
 wǒ yìbān/tōngcháng **dōu** chéng gōngjiāochē shàngxué.
 I usually DOU take bus go.to.school
 ‘I usually take the bus to school.’

In some cases, ambiguity could arise between the quantifier-distributor use of *dōu* and the situational genericity-regarding use of *dōu*:

- (70) 小艾 和 小白 一天 都 接 十个 电话.
 Xiǎoài hé Xiǎobái yī-tiān **dōu** jiē shí-gè diànhuà.
 Alice and Bob one-CLF.day DOU receive ten-CLF phone.call
 ‘Alice and Bob BOTH receive ten phone calls a day.’
 OR
 ‘Alice and Bob ALWAYS receive ten phone calls a day.’

As *dōu* is located just before the verb no matter what focus it takes, this sentence is ambiguous between the meanings of ‘Alice and Bob BOTH receive ten calls a day’ and ‘Alice and Bob receive ten calls a day ALL THE TIME’. Only the former sense of (70) is true in a situation where Alice and Bob each receive ten phone calls a day on most of the days, but not on a small number of exceptional days; meanwhile, only the latter sense of (70) is true in a situation where on every single day, Alice receives seven calls and Bob receives three calls.

As for theoretical explanations for this phenomenon, Krifka et al. 1995 and Eckardt 1999 offer semantics for generic sentences. Roughly speaking, Krifka et al. 1995 advocates for a theory where the truth value of a generic sentence is evaluated in some ideal normal world, and Eckardt 1999 endorses an approach that computes the truth value of a generic sentence with prototypical, canonical individuals of a class of objects. Nonetheless, both studies propose some form of a generic operator GEN that takes arguments constituting the prejacent. It is plausible that a hybrid between the semantics of alternatives and generic sentences could account for the data above, but I leave an account for this occurrence and the consequences for the semantics of generics and *dōu* for further research.

5.5 Summary

In this section, I have identified that focus alternatives can be sets of situations and predicted the part-time NPI behaviour of *zuò mèng* through a theory of idiomatic situational foci. A similar idea could be applied to the semantics of generic constructions, warranting further research on similar phenomena in other languages.

6 Conclusion

In this dissertation, I have pursued a unified semantics of *yě* and *cái* in parallel with that of *dōu* that accounts for their multiple functions: they are pre-exhaustification exhaustifiers operating on orthogonal alternatives. I have allocated the temporal use of *cái* and the *also*-like use of *yě* as their primary uses, mirroring the quantifier-distributor use of *dōu*; the scalar operator uses of *yě* and *cái* are categorised as secondary uses that are derived from the primary use through a semantic weakening operation on orthogonal alternatives borrowed from that of *dōu*. The lack of orthogonal alternatives in the D-alternatives leads to *yě* and *cái*'s inability to license FCIs under three-valued Kleene logic.

The scalar operator uses of *dōu*, *yě* (and by extension, *cái*) are not restricted to alternatives of individuals, scales and disjunctions, but also situations. Idiomatic situational foci is a particular case of situational foci that acquires a superlative effect in suitable pragmatic conditions due to the interaction between the core lexical meaning and the likelihood order relation intrinsic to *dōu*. Such a theory, applied to *zuò mèng*, explains why *zuò mèng* displays negative polarity behaviour only on some occasions: the choice of attitude verbs has a pragmatic restriction on the polarity of superlatives that is sensible in the context.

This dissertation has pinpointed some imperfections in the theory of Mandarin Chinese *wh*-items, underlined the association between logic and semantic composition and offered some outlook for the theory of generic constructions with respect to *dōu*, all of which may be of interest for further research.

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